

This image illustrates the difference between bitmap and vector images. The vector image can be scaled indefinitely without loss of image quality, while the bitmap cannot.

SVG allows three types of graphic objects:

- Vector Graphics
- Raster Graphics
- Text

Graphical objects can be grouped, styled, transformed and composited into previously rendered objects. SVG does not directly support z-indices that separate drawing order from document order for objects, which is a drawback with respect to other vector mark-up languages like VML. Text can be in any XML namespace suitable to the application, which enhances searchability and accessibility of the SVG graphics. The feature set includes nested transformations, clipping paths, alpha masks, filter effects, template objects and extensibility.

While being primarily designated as a vector graphics mark-up language, the specification is also designed with the basic capabilities of a page description language, similar to Adobe's PDF. It contains provisions for rich graphics, and is also compatible with the CSS specification's properties for styling purposes. Therefore, unlike XHTML and XSL-FO which are layout-oriented languages, SVG is a fully presentation-oriented language. A much more print-specialized subset of SVG known as SVG Print (authored by Canon, HP, Adobe and Corel) is currently a W3C working draft.

SVG drawings can be dynamic and interactive. Time-based modifications to the elements can be described in SMIL, or can be programmed in a scripting language (e.g., ECMAScript). The W3C explicitly recommends SMIL as the standard for animation in SVG. However, it is common to find SVG animated with ECMAScript because it is a language that many developers already understand, and is more compatible with existing renderers. A rich set of event